Test-Time Augmentation-Enhanced Knowledge Distillation for Brain Tumor MRI Classification

Abstract—**Deep learning models for brain tumor MRI classification often face a trade-off between accuracy and efficiency. In this paper, we propose a Test-Time Augmentation (TTA)-enhanced knowledge distillation framework to improve the stability of teacher predictions. By aggregating multiple augmented views, TTA reduces prediction variance and provides more reliable supervision signals for distillation. The proposed approach transfers knowledge from a CNN ensemble to lightweight Hybrid Vision Transformers, achieving up to $45\times$ model compression while maintaining competitive performance. These results demonstrate the effectiveness of TTA-enhanced distillation for efficient medical imaging applications.**

Index Terms—**Knowledge Distillation, Test-Time Augmentation, Hybrid Vision Transformers, Model Compression, CNN Ensembles, Medical Image Classification.**

I. Introduction

Deep learning has significantly advanced brain tumor classification in medical imaging [1], yet balancing accuracy and computational efficiency remains a critical challenge for real-world clinical deployment [2]. High-performance systems often rely on large CNN ensembles, which, despite their accuracy, are computationally expensive and unsuitable for resource-constrained environments such as edge devices or hospital systems with limited infrastructure.

Vision Transformers (ViTs) have demonstrated strong capability in capturing global dependencies [3], but they typically require large-scale data and incur high computational costs. Meanwhile, Hybrid Vision Transformers such as MobileViT [4] and EdgeNeXt [5] have been proposed, combining CNN inductive biases with Transformer-based global modeling [6]. However, these lightweight models still underperform compared to large CNN ensembles.

Knowledge Distillation (KD) offers an effective paradigm for transferring knowledge from high-capacity teachers to compact students [7]. Nevertheless, conventional KD approaches often rely on single teachers, limiting knowledge diversity [8], and may suffer from unstable or noisy soft targets that degrade distillation performance [9].

Despite recent progress, existing approaches still exhibit key limitations. Most knowledge distillation methods rely on single teachers or unstable soft targets, while ensemble-based teachers are rarely combined with Test-Time Augmentation (TTA) [10] to improve supervision stability. Moreover, lightweight Hybrid Vision Transformers remain underexplored as students under such teacher settings, motivating a more effective distillation framework for brain tumor classification tasks.

In this work, we reinterpret Test-Time Augmentation (TTA) not merely as an inference-time performance enhancement technique, but as a mechanism for reducing prediction variance and stabilizing teacher supervision signals, particularly in the context of brain tumor classification from MRI data. Due to variations in imaging conditions, patient anatomy, and tumor appearance, MRI-based predictions are often sensitive to small input perturbations, leading to unstable soft targets for knowledge distillation. By aggregating predictions across multiple augmented views, TTA approximates an expectation over input perturbations, resulting in smoother and more reliable soft targets for distillation.

Building upon this insight, we propose a knowledge distillation framework that leverages TTA-enhanced CNN ensembles as a robust teacher to train lightweight Hybrid Vision Transformers, including MobileViT-XXS and EdgeNeXt-XXS, for efficient brain tumor classification. The proposed framework improves the stability of knowledge transfer while maintaining strong parameter efficiency, making it suitable for deployment in resource-constrained medical imaging scenarios.

The **main contributions** of this work are as follows:

1) **Variance-stabilized teacher.** We propose a TTA-enhanced CNN ensemble as a variance-reduced teacher to generate more reliable soft targets for brain tumor MRI classification.
2) **Efficient distillation to Hybrid Vision Transformers.** We develop a knowledge distillation framework for MobileViT-XXS and EdgeNeXt-XXS, enabling strong accuracy with high efficiency.
3) **High compression with competitive performance.** The proposed method achieves up to $45\times$ model compression while maintaining performance comparable to the teacher.

II. Related Work

A. CNN-based Architectures for Brain Tumor Classification

Deep Convolutional Neural Networks (CNNs) such as ResNet, DenseNet, and VGG have been widely adopted for brain tumor classification due to their strong feature extraction capabilities [11]–[13]. To further improve performance, ensemble learning is often employed to enhance generalization and reduce variance [14]. Recent studies have proposed various enhancements to CNN-based models, including residual-based architectures [15], attention mechanisms such as CBAM [16], and hybrid designs combining multiple CNN backbones [17]. Additionally, lightweight CNNs with image enhancement techniques have been explored to improve efficiency while maintaining competitive performance [18]. Despite these advances, CNN-based approaches—especially ensembles—often incur high computational cost, limiting their applicability in resource-constrained settings.

B. Evolution of Hybrid Vision Transformers

Vision Transformers (ViTs) have introduced a paradigm shift by modeling long-range dependencies through self-attention mechanisms [19]. However, their lack of inductive bias and high data requirements hinder their effectiveness in medical imaging tasks. To address these limitations, Hybrid Vision Transformers such as MobileViT [4] and EdgeNeXt [5] integrate CNN-based local feature extraction with Transformer-based global context modeling. Recent works have demonstrated their effectiveness for brain tumor classification in resource-constrained or on-device scenarios [20]. Nevertheless, these models still exhibit a performance gap compared to large CNN ensembles.

C. Knowledge Distillation (KD) Paradigms

Knowledge Distillation (KD) is a widely used model compression technique that transfers knowledge from large teacher models to smaller student models [7]. Existing KD approaches can be broadly categorized into response-based, feature-based, and relation-based methods. In medical imaging, advanced KD techniques have been proposed to enhance feature representation and generalization, including coordinate-based distillation [21] and dynamic weighted KD strategies [22]. Multi-teacher distillation has also been explored to improve knowledge diversity and efficiency in brain tumor classification [23]. However, effectively leveraging ensemble teachers while ensuring stable and informative soft targets remains a challenge. In particular, the integration of Test-Time Augmentation (TTA) into the distillation process, along with systematic optimization for Hybrid Vision Transformer students, is still underexplored. This work addresses this gap by proposing a robust KD framework that combines ensemble learning, TTA, and optimized distillation.

III. Methodology

The proposed framework consists of three stages: (1) constructing a teacher via CNN ensembling with Test-Time Augmentation (TTA), (2) defining lightweight Hybrid Vision Transformer students, and (3) optimizing knowledge transfer through logit-based distillation. An overview of the proposed knowledge distillation framework is illustrated in Figure 1.

A. Robust Teacher Construction

We establish a candidate pool of heterogeneous CNN architectures comprising DenseNet121 [24], ResNet50 [25], VGG19 [26], and MobileNetV2 [27]. To construct the optimal teacher, we empirically select the best-performing subset of these models tailored to each specific dataset. All models are fully fine-tuned by replacing their classification heads with a unified lightweight head (AdaptiveAvgPool2d, Flatten, Dropout with a rate of 0.3, and a Linear output layer) and training using Focal Loss. At inference time, TTA is applied by generating N augmented samples $\{x'_j\}$ for each input x. The final teacher logits z_t are computed as:

$$z_t = \frac{1}{M(N+1)} \sum_{i=1}^{M} \left(f_i(x) + \sum_{j=1}^{N} f_i(x'_j) \right)$$

where M is the number of models and f_i denotes the i-th CNN model. This formulation stabilizes predictions and reduces uncertainty.

B. Student Models

We adopt MobileViT-XXS and EdgeNeXt-XXS as lightweight students, which integrate CNN-based local feature extraction with Transformer-based global context modeling. Both models use the same classification head design as the teacher.

C. Optimized Knowledge Distillation Framework

We employ a hybrid objective that integrates supervised learning with teacher guidance:

$$L_{total} = (1-\alpha)L_{FL} + \alpha T^2 L_{KL}$$

This objective balances Focal Loss (L_{FL}) [28] and Kullback–Leibler divergence (L_{KL}) via α, enabling the student to learn from both hard labels and teacher's soft logits. The scaling factor T^2 ensures consistent gradient magnitudes between the two components.

1) Supervised Component: An unweighted Focal Loss is adopted to dynamically scale the cross-entropy based on prediction confidence, heavily emphasizing hard-to-classify samples:

$$L_{FL}(p_t) = -(1-p_t)^\gamma \log(p_t)$$

where p_t denotes the predicted probability for the target class.

2) Distillation Component: Knowledge transfer is achieved by minimizing L_{KL} between the softened outputs of the teacher and student. A temperature $T > 1$ produces smoother probability distributions, revealing inter-class relationships that improve generalization.

3) Hyperparameter Optimization: We perform grid search on validation sets from 5-fold cross-validation to determine optimal T and α, selecting configurations that maximize performance while ensuring stable convergence.

IV. Experiments

A. Experiment Setup

1) Datasets and Pre-processing: We evaluate our framework on two public brain tumor MRI datasets: Figshare [29] (3,064 images; 3 classes: glioma, meningioma, pituitary; evaluated under 5-fold cross-validation (5-FCV), 8:2, and 8:1:1 splits) and Kaggle Brain Tumor MRI Dataset (BT-MRI) [30] (7,023 images; 4 classes: glioma, meningioma, pituitary, and no tumor). The 5-FCV setting follows the official patient-level partition to avoid data leakage, while the other splits are performed at the image-level for comparison with prior

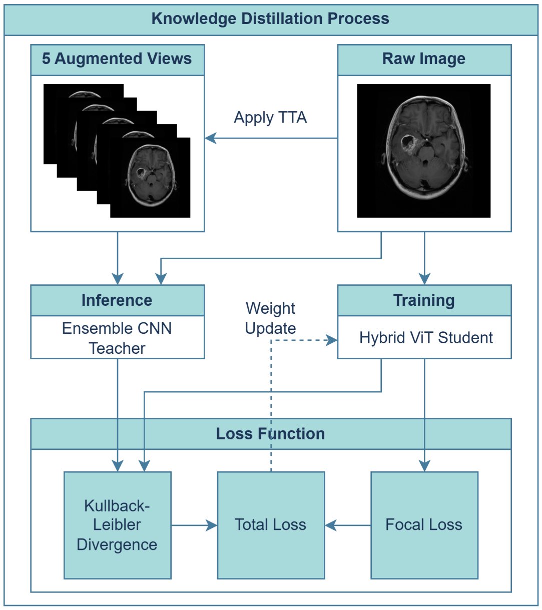

Fig. 1. Overview of the proposed knowledge distillation framework.

work. We therefore consider 5-FCV as the primary indicator of generalization.

During training, all images are resized to 256×256 and augmented using random horizontal flipping ($p = 0.5$), rotation ($\pm 36°$), affine scaling (0.9–$1.1\times$), and color jittering for brightness and contrast ($\pm 10\%$). A 224×224 center crop is then applied, followed by ImageNet normalization. For Test-Time Augmentation (TTA), we generate multiple views per input image using horizontal flipping ($p = 0.5$) and mild color jittering (brightness $\pm 10\%$, contrast 0.9–1.1). These lightweight transformations preserve distribution consistency while improving prediction robustness.

2) Implementation Details: Experiments are implemented in PyTorch 2.10.0 and conducted on an NVIDIA P100 GPU, with a fixed random seed of 24520152. We use the Adam optimizer with an initial learning rate of 1×10^{-4} and a ReduceLROnPlateau scheduler (factor 0.2, patience 3, minimum 10^{-7}). Early stopping is applied with a patience of 6 epochs for individual models and 15 epochs during distillation. All models are trained using unweighted Focal Loss ($\gamma = 2$).

Hyperparameters are selected via grid search on 5-FCV validation sets, with $T \in \{2, 3\}$ and $\alpha \in \{0.4, 0.5, 0.6\}$. The best configurations are $T = 3, \alpha = 0.5$ for MobileViT-XXS and $T = 3, \alpha = 0.6$ for EdgeNeXt-XXS, which are used in all subsequent experiments, except during cross-architecture benchmarking where a unified setting is applied. For Test-Time Augmentation (TTA), $N = 5$ augmented views are generated per image.

3) Evaluation Metrics: Performance is evaluated using accuracy, precision, recall (sensitivity), specificity, and F1-score. We also report the total number of model parameters and the compression ratio, defined as the total parameters of the specific teacher ensemble divided by the parameters of the distilled student model. For example, as shown in Table I for the BT-MRI split, a teacher ensemble with $\sim$43.54M parameters and a MobileViT-XXS student with $\sim$0.95M parameters yield a compression ratio of 45.72$\times$.

V. Results and Discussion

A. Performance of Teacher and Distilled Students

Distilled Students: As shown in Table I, the distilled students achieve remarkable parameter efficiency, operating with up to $\sim$45$\times$ fewer parameters than the teacher ensemble. The compression ratio varies across different splits due to differences in the selected teacher ensembles and their parameter sizes. Despite this massive compression, knowledge distillation generally enhances the performance of lightweight students.

The most substantial gain is observed on the 5-FCV split, where MobileViT-XXS improves from 94.81% to 96.92% (+2.11%). On certain splits (Figshare 8:2 and 8:1:1), the distilled MobileViT-XXS slightly outperforms the TTA-enhanced teacher ensemble by 0.33%. This phenomenon can be attributed to the regularization effect of knowledge distillation, potentially leading to improved generalization. However, the benefit of distillation may depend on the data partitioning strategy.

TABLE I
Comparison of Distilled Students vs. Baseline Students. Positive Δ values indicate the accuracy gain achieved through distillation.

Dataset	Student Model	Comp. Ratio	Base Acc	Distilled Acc	Delta Student
BT-MRI	MobileViT-XXS	45.72$\times$	99.54%	**99.77%**	+0.23%
	EdgeNeXt-XXS	37.57$\times$	99.69%	99.62%	-0.07%
Figshare 5-FCV	MobileViT-XXS	34.35$\times$	94.81%	96.92%	**+2.11%**
	EdgeNeXt-XXS	28.22$\times$	95.72%	**97.06%**	+1.34%
Figshare 8:2	MobileViT-XXS	9.65$\times$	97.55%	**98.86%**†	+1.31%
	EdgeNeXt-XXS	7.93$\times$	97.55%	98.21%	+0.66%
Figshare 8:1:1	MobileViT-XXS	45.74$\times$	98.05%	**98.70%**†	+0.65%
	EdgeNeXt-XXS	37.58$\times$	96.09%	97.39%	+1.30%

† Indicates distilled student outperformed the TTA-enhanced teacher by 0.33%.

TTA-enhanced Teacher Ensemble: To establish a strong teacher for distillation, we compare standalone CNNs with ensemble and TTA-enhanced variants (Figure 2). While individual models provide strong baselines (e.g., 97.00% on Figshare 5-FCV and 99.77% on BT-MRI), ensemble learning generally improves performance. Applying TTA further increases accuracy in most cases, yielding peak accuracies of 99.85% on BT-MRI and 97.21% on Figshare (5-FCV). However, on the Figshare 8:2 split, TTA slightly degrades performance, which may be attributed to over-smoothing effects when predictions are already highly confident. These results suggest that TTA tends to be more beneficial in challenging settings, while its impact may diminish in relatively saturated scenarios.

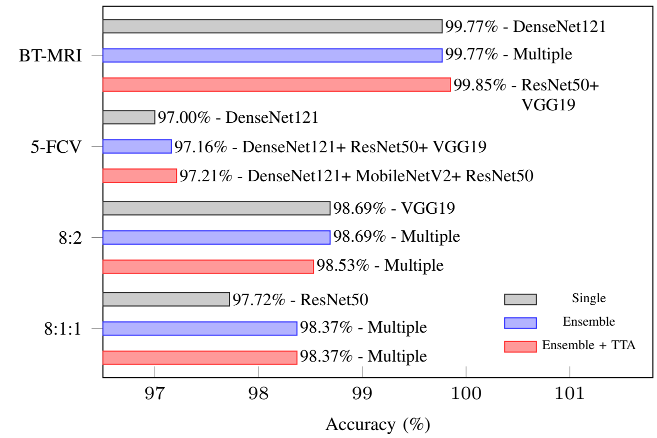

Fig. 2. Accuracy comparison of single models, baseline ensembles, and TTA-enhanced ensembles across different datasets and splits.

B. Statistical Significance Analysis

We perform paired t-tests only on the 5-FCV results, as the five independent folds ($n = 5$, $df = 4$) provide the variance needed for statistical testing, unlike a single hold-out split.

For the null hypothesis that the distilled student and the TTA-enhanced teacher have equal mean accuracy, both EdgeNeXt-XXS and MobileViT-XXS fail to reject H_0 ($t = 1.0046$, $p = 0.3719$, 95% CI: $[-0.26\%, 0.56\%]$; $t = 1.2317$, $p = 0.2855$, 95% CI: $[-0.37\%, 0.96\%]$), indicating no statistically significant difference.

For the directional hypothesis that distillation improves accuracy over the baseline student, both models show significant gains ($t = 3.9357$, $p = 0.00851$; $t = 6.1453$, $p = 0.00178$), with 95% CIs of $[0.40\%, 2.30\%]$ and $[1.15\%, 3.06\%]$, respectively. These results confirm consistent improvement over independently trained students.

C. Comparison with State-of-the-Art Methods and Distilled Lightweight CNNs

Benchmark against State-of-the-Art Methods: We benchmark the distilled Hybrid ViTs against state-of-the-art (SOTA) methods that report parameter counts on the Figshare and BT-MRI datasets (Table II). By leveraging high-fidelity soft targets, the proposed models achieve competitive accuracy with a much smaller parameter footprint. For example, the distilled MobileViT-XXS (0.95M) outperforms a 79.3M-parameter Vision Differential Transformer on BT-MRI (99.77% vs. 99.31%), and remains competitive against larger 53M-parameter models on Figshare. Overall, the results show that our method offers strong parameter efficiency, matching or exceeding much larger models at a fraction of the cost.

Benchmark against Distilled Lightweight CNNs: We further compare the Hybrid ViTs against five lightweight CNN architectures: ShuffleNetV2 (x1.0) [34], GhostNet [35], MobileNetV3-Small [36], MobileNetV4-Conv-Small [37], and SqueezeNet 1.0 [38]. All models (both the Hybrid ViTs and the lightweight CNNs) are distilled using a unified configuration ($T = 3$, $\alpha = 0.5$) to ensure a fair and controlled comparison across different architectures, and evaluated across all dataset splits, with results reported as mean accuracy.

TABLE II
COMPARISON OF THE PROPOSED DISTILLED MODELS WITH STATE-OF-THE-ART METHODS.

Study	Method	Params	Accuracy
Celik et al. (BT-MRI) [31]	Vision Differential Transformer	$\sim$79.3M	99.31%
Nassar et al. (Figshare 7:3) [32]	Hybrid Deep Learning NASNet-Mobile	53M	99.31%
Krishnan et al. (BT-MRI) [19]	Rotation Invariant Vision Transformer	7.5M	98.60%
Nassar et al. (Figshare 5-FCV) [32]	Hybrid Deep Learning NASNet-Mobile	53M	98.3%
Alim et al. (BT-MRI) [23]	Multi-teacher KD + Ensembling (Tiny CNNs)	0.23M	98.02%
Hammad et al. (Figshare 7:1:2) [33]	Lightweight End-to-End Deep Learning Model	2.5M	96.86%
Ours (BT-MRI)	**Distilled MobileViT-XXS**	**0.95M**	**99.77%**
Ours (Figshare 8:2)	**Distilled MobileViT-XXS**	**0.95M**	**98.86%**
Ours (Figshare 5-FCV)	**Distilled EdgeNeXt-XXS**	**1.16M**	**97.06%**

As shown in Figure 3, the proposed models achieve a favorable size–performance trade-off. MobileViT-XXS attains the highest mean accuracy (98.56%) with $\sim$0.95M parameters, while EdgeNeXt-XXS also remains competitive (98.07%, $\sim$1.16M). In comparison, GhostNet reaches 98.02% with $\sim$3.91M parameters, and MobileNet variants show slightly lower accuracy with larger model sizes. Although SqueezeNet 1.0 is the most compact ($\sim$0.74M), it yields the lowest accuracy (95.32%). These results indicate that the proposed Hybrid ViTs offer favorable parameter-to-performance efficiency compared to conventional lightweight CNNs.

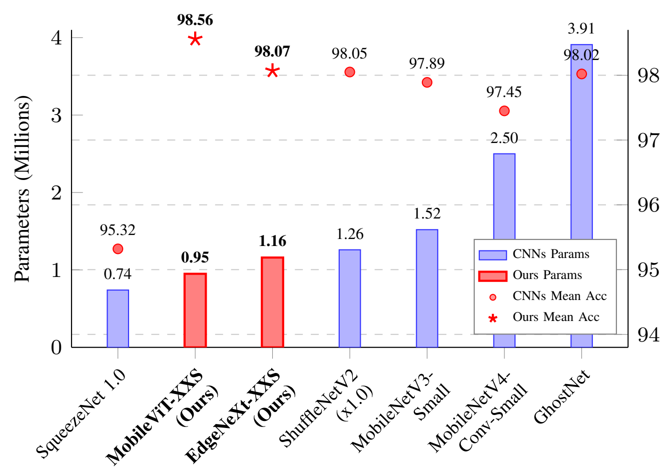

Fig. 3. Performance and parameter comparison of distilled Hybrid ViTs against distilled lightweight CNNs across datasets. Our proposed models are highlighted.

D. Comprehensive Evaluation and Ablation Study

Table III reports extended classification metrics and an ablation study on the effect of Test-Time Augmentation (TTA).

The results suggest that TTA is more beneficial on challenging splits, particularly Figshare 5-FCV. For instance, MobileViT-XXS improves its F1-score from 95.72% (w/o TTA) to 96.41% (w/ TTA), indicating improved teacher supervision through reduced prediction uncertainty.

In contrast, performance on BT-MRI remains saturated at approximately 99.7% across configurations, suggesting a ceiling effect where additional augmentation has limited impact. Moreover, some cases exhibit mixed trends across evaluation metrics, indicating that improvements in accuracy do not always translate uniformly to precision, recall, and F1-score. This may be attributed to the different sensitivities of evaluation metrics to changes in decision boundaries induced by TTA and distillation, even under Focal Loss optimization.

Overall, these results suggest that the effect of TTA depends on data complexity, with more noticeable gains on challenging splits (e.g., Figshare 5-FCV) and marginal impact on saturated datasets such as BT-MRI.

VI. Conclusion

In this paper, we proposed a knowledge distillation framework that leverages TTA-enhanced CNN ensembles as a variance-stabilized teacher for brain tumor classification from MRI data. By reinterpreting Test-Time Augmentation as a mechanism for reducing prediction variance, the proposed approach generates more reliable soft targets, improving the effectiveness of knowledge transfer to lightweight Hybrid Vision Transformers.

Experimental results demonstrate that the distilled models achieve competitive performance while reducing model size by up to $45\times$, highlighting a strong accuracy–efficiency trade-off. These results suggest that variance-aware distillation is a promising direction for developing efficient and robust models in medical imaging.

Future work will explore adaptive TTA strategies and extend the framework to more complex settings, including 3D medical imaging and cross-dataset generalization.

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TABLE III
COMPREHENSIVE EVALUATION METRICS AND TTA ABLATION STUDY ACROSS DIVERSE DATA SPLITS

Dataset Split	Model Configuration	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)
Figshare 5-FCV	Baseline MobileViT-XXS	94.81	92.20	92.25	96.65	92.09
	Distilled MobileViT-XXS (w/o TTA)	96.28	95.88	95.58	98.11	95.72
	Distilled MobileViT-XXS (w/ TTA)	**96.92**	**96.62**	**96.24**	**98.39**	**96.41**
	Baseline EdgeNeXt-XXS	95.72	93.78	93.67	97.31	93.60
	Distilled EdgeNeXt-XXS (w/o TTA)	96.64	96.13	96.18	98.34	96.15
	Distilled EdgeNeXt-XXS (w/ TTA)	**97.06**	**96.61**	**96.56**	**98.53**	**96.56**
Figshare 8:2	Baseline MobileViT-XXS	97.55	97.56	96.82	98.67	97.17
	Distilled MobileViT-XXS (w/o TTA)	98.53	98.41	98.30	99.23	98.35
	Distilled MobileViT-XXS (w/ TTA)	**98.86**	**97.87**	**97.18**	**99.36**	**97.52**
	Baseline EdgeNeXt-XXS	97.55	97.70	96.77	98.66	97.18
	Distilled EdgeNeXt-XXS (w/o TTA)	**98.37**	**98.12**	**98.12**	**99.19**	**98.12**
	Distilled EdgeNeXt-XXS (w/ TTA)	98.21	97.95	98.00	99.09	97.98
Figshare 8:1:1	Baseline MobileViT-XXS	98.05	98.11	97.66	98.94	97.87
	Distilled MobileViT-XXS (w/o TTA)	97.72	97.62	97.19	98.80	97.39
	Distilled MobileViT-XXS (w/ TTA)	**98.70**	**98.83**	**98.36**	**99.25**	**98.58**
	Baseline EdgeNeXt-XXS	96.09	96.21	94.84	97.87	95.39
	Distilled EdgeNeXt-XXS (w/o TTA)	**98.05**	**98.28**	**97.18**	**98.92**	**97.65**
	Distilled EdgeNeXt-XXS (w/ TTA)	97.39	97.15	96.83	98.66	96.98
BT-MRI	Baseline MobileViT-XXS	99.54	99.51	99.50	99.85	99.51
	Distilled MobileViT-XXS (w/o TTA)	**99.77**	**99.76**	**99.75**	**99.93**	**99.75**
	Distilled MobileViT-XXS (w/ TTA)	**99.77**	**99.76**	**99.75**	**99.93**	**99.75**
	Baseline EdgeNeXt-XXS	99.69	99.67	99.67	99.90	99.67
	Distilled EdgeNeXt-XXS (w/o TTA)	**99.77**	**99.75**	**99.75**	**99.93**	**99.75**
	Distilled EdgeNeXt-XXS (w/ TTA)	99.62	99.63	99.59	99.87	99.61

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